

Table I — Architecture parameters (864-PE configuration)

Parameter	Value
PE array (M × N)	9 × 96 = 864 PEs
Data precision	Q8.8 fixed-point (16-bit)
Simulation clock (fclk)	100 MHz (behavioural)
KPU pipeline warmup	Z(16) + N_PE(96) + 4 = 116 cycles
Test: Conv(nl=16) + FC(nl=5, N=5)	2-layer CNN, class=4 [PASS]

Table II — Latency breakdown

Stage	Cycles / Time
Total inference latency	2660 ns / 266 cycles
Active COMPUTE cycles	253 of 266 (95.1 %)
Overhead cycles	13 of 266 (4.9 %)
KPU warmup cycles (pipeline flush)	116 of 266 (included in COMPUTE)
CU classification (CLASSIFY state)	5 cycles (N_classes=5)

Table III — Throughput and TOPS ($\Theta T = 2 \times NPE \times fclk \times \Omega \times \sigma$)

Metric	Value
NPE (M=9 × N=96)	864
Ω (PE efficiency)	1.00 (1×1 / 3×3 kernels)
σ (time efficiency, measured)	0.9511 (95.1 %)
Frame rate (sim @ 100 MHz)	375,939 fps
ΘT @ 100 MHz, 864 PE (sim)	0.1644 TOPS
ΘT @ 1.25 GHz, 864 PE (FPGA)	2.055 TOPS ($\sigma=0.951$)
ΘT @ 3.85 GHz, 864 PE (ASIC, $\sigma=1$)	6.6528 TOPS (paper: 6.65 TOPS ✓)

Table IV — 54 PE vs 864 PE comparison @ 100 MHz

Metric	54 PE	864 PE
ΘT (100 MHz)	0.00867 TOPS	0.16436 TOPS